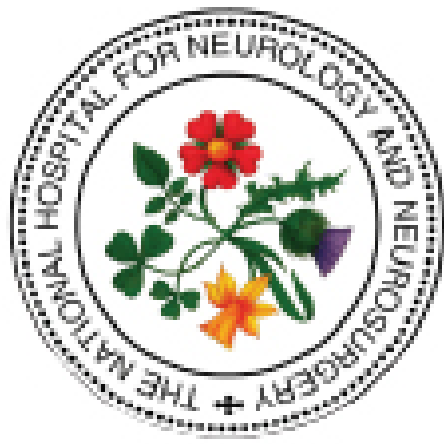




1. DIAGNOSING DEATH by NEUROLOGICAL CRITERIA (DNC)

A Code of Practice for the Diagnosis and Confirmation of Death in UK 2025
Academy of Medical Royal Colleges (AoMRC)

Code of Practice for Diagnosis and Confirmation of Death

'When the injury results in permanent cessation of brainstem function, a state beyond coma, even if circulation and other organ functions can be artificially maintained successfully, death of the person has occurred.'

In nearly all circumstances death is a process rather than an event, with different organs, tissues and cells ceasing function and beginning decay at different rates. It is therefore necessary to define a point in this process where death can be diagnosed and confirmed in an accurate, standardised and timely manner.

While death itself has not changed since the last Academy Code was published in 2008, medical practice and technologies have advanced, particularly in the fields of resuscitation, intensive care, neurosurgery and organ donation. These factors led to the need to provide updated guidance for all those involved in the diagnosis and confirmation of death.

In the UK a biomedical definition of death, based on loss of function in the brainstem, has been medically accepted since 1979 and, although there is no statutory definition of death in the UK, the courts have recognised the concept of brainstem death as legal death. The 2025 Code includes several important updates and incorporates global medical consensus and lessons learned from individual clinical cases internationally.

The primary purpose of the Code is to outline the criteria by which death can be diagnosed and confirmed whatever the circumstance in which the death has occurred.

Three sets of diagnostic criteria can be used to confirm death and the clinical circumstances where a particular set of criteria would be most appropriate to use are outlined in the updated Code:

- Somatic criteria** are appropriate to use when death follows overwhelming physical trauma or when death is suspected to have occurred a considerable time before. These criteria can be used by healthcare professionals or other competent individuals.
- Circulatory criteria** are appropriate to use when death follows cardiorespiratory arrest and is the most common way death is diagnosed and confirmed. These criteria should only be used by healthcare professionals.
- Neurological criteria** are appropriate to use when death follows a devastating brain injury but where the circulation is being maintained by ongoing mechanical ventilation and other intensive care interventions. Approximately 1,500 patients each year in the UK have their death diagnosed and confirmed using neurological criteria. These criteria should only be used by doctors.

Neurological criteria do not require confirmation of anatomical destruction of the brainstem or the cessation of all neurological or other bodily functions. What does follow from a diagnosis of death using neurological criteria is that there is sufficient severity of brain injury that there can be no form of consciousness associated with human life, such as: arousal, wakefulness, and the ability to feel, to be aware of, or to have subjective experience.

IMPORTANT UPDATES in 2025 CODE for DIAGNOSING DEATH BY NEUROLOGICAL CRITERIA	
AGE CATEGORIES	Children 2 years and above - criteria as per adults <i>For children younger than 2-years, additional diagnostic caveats are required</i>
APNOEA TEST	START: PaCO ₂ ≥5.3kPa TIME: Observed for minimum 5 minutes TARGET: PaCO ₂ must be at least 8.0kPa END: PaCO ₂ 2.7kPa > START PaCO ₂
TIME OF DEATH	Completion of 2nd set of clinical tests (or timing of ancillary testing if this occurs after clinical tests)
MINIMUM TEMPERATURE	Minimum core temperature 36°C
BOTH EYES / BOTH EARS MUST BE EXAMINABLE	If not able to examine both ears / both eyes, ancillary investigation required

January 2025

Academy of Medical Royal Colleges

A Code of Practice for the diagnosis and confirmation of death
2025 Update

<https://www.aomrc.org.uk/publication/a-code-of-practice-for-the-diagnosis-and-confirmation-of-death-2025-update/>

Form for the Diagnosis of Death using Neurological Criteria

This form should be used in conjunction with the AoMRC 2025 Update A Code of Practice for the Diagnosis and Confirmation of Death (Code).

The latest version of this form must be used – found at www.ficm.ac.uk – including if the form has been transcribed for use into an electronic health record.

HOSPITAL ADDRESSOGRAPH or
Surname
First Name
Date of Birth
NHS / CHI Number

Preparation

1. Patient selection

- A patient following devastating brain injury who remain deeply comatose (GCS 3/15), have no observed brainstem reflexes and are apnoeic with their lungs being mechanically ventilated, but in whom circulation and other bodily functions persist. (Code 4.2)
- This form is applicable for use in adults and children older than 2-years corrected age post term. For children younger than 2-years, additional diagnostic caveats are required – use the Children 37-weeks to 2-years Testing Form. (Code 4.3b, Appendix 2)
- Families should ideally be offered the opportunity to observe a set of clinical tests to confirm death. Often, the second set of tests is the most appropriate and useful for families to witness. (Code 6.9)
- Patients requiring extracorporeal membrane oxygenation (ECMO): Guidance to supplement the 2008 Code for patients on ECMO has been previously published and remains valid.

2. Ancillary investigations (Page 5, Code 4.54-4.67, Appendix 2, A9)

end-tidal CO₂; peripheral nerve stimulator; thermometer; patient warming device.
Clinical testing of brainstem reflexes: Bright light source and/or pupillometer; small gauze sterile swabs; otoscope with disposable earpieces, ice-cold water, 50 ml syringe, disposable quill (or equivalent) if needed; tongue depressor or firm suction catheter (e.g. Yankauer sucker); laryngoscope or video laryngoscope; endotracheal suction catheter.
Apnoea test: Arterial blood gas analysis including at least 4 blood gas syringes, CPAP circuit (e.g. Mapleson C or equivalent).

Adults and Children over 2-years, January 2025

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The use of Cerebral CT Angiography as an Ancillary Investigation to support a clinical Diagnosis of Death using Neurological Criteria

<https://www.ficm.ac.uk/the-use-of-cerebral-ct-angiography-as-an-ancillary-investigation-to-support-a-clinical-diagnosis-of-death-using-neurological-criteria/>